Calificativ (Săptămâna 7) – Exerciții Legea lui Demeter & Principii S.O.L.I.D.

I. Legea lui Demeter

1. Următoarele fragmente de cod încalcă Legea lui Demeter? Argumentați răspunsul. Dacă este încălcată, oferiți o soluție pentru a respecta Legea lui Demeter.

a.

```
public void printSomePoints(List<Point> points) {

   int printedPoints = 0;
   for (Point point : points) {
       if (someCondition()) {
           System.out.println(String.format("Point %d (%f,%f)",
                   printedPoints, point.getX(), point.getY()));
       }
   }
}

class Point {
   private double x;
   private double y;

   //getters and setters
}
```

b.

```
public void printDistanceBetweenRandomPointsOfTwoSquares(Square
s1, Square s2) {
   Point randomPoint1 = s1.getRandomPoint();
   Point randomPoint2 = s2.getRandomPoint();

   System.out.println(randomPoint1.distanceTo(randomPoint2));
}

@AllArgsConstructor
class Point {
   @Getter private double x;
   @Getter private double y;

   public double distanceTo(Point p){
       return Math.hypot(x-p.x, y-p.y);
   }
}
```

```java
@Data //getters and setters
@AllArgsConstructor
class Square {
   private Point[] points = new Point[4];

   public Point getRandomPoint(){
       int index = (int) (Math.random()*4);
       return points[index];
   }
}
```

c.

```java
public void someMethod(Student student) {
   Faculty faculty = student.getFaculty();

   System.out.println(faculty.getName());
   System.out.println(faculty.getStudentNo());
}

@Data
@AllArgsConstructor
class Student {
   private String name;
   private Faculty faculty;

}

@Data
@AllArgsConstructor
class Faculty {
   private List<Student> students;

   public int getStudentNo() { return students.size(); }
   public void removeStudent(Student student) {
students.remove(student); }
   public void addStudent(Student student) {
students.add(student); }
}
```

d.

```java
public void someMethod() {
   //...some code
   Student student = Student.builder()
           .lastName("LN")
           .firstName("FN")
           .age(20)
           .bestFriend(null)
```

```
            .build();
    System.out.println(student);
    //...some other code
}

@AllArgsConstructor
class Student {
    private String firstName;
    private String lastName;
    private Integer age;
    private Student bestFriend;

    public static StudentBuilder builder() {
        return new StudentBuilder();
    }
}

class StudentBuilder {
    private String firstName;
    private String lastName;
    private Integer age;
    private Student bestFriend;

    StudentBuilder() { }

    public StudentBuilder firstName(String firstName) {
this.firstName = firstName; return this; }

    public StudentBuilder lastName(String lastName) {
this.lastName = lastName; return this; }

    public StudentBuilder age(Integer age) { this.age = age;
return this; }

    public StudentBuilder bestFriend(Student bestFriend) {
this.bestFriend = bestFriend; return this; }

    public Student build() { return new Student(firstName,
lastName, age, bestFriend); }
}
```

2. Care este diferența dintre o structură de date și o clasă? La care dintre ele se aplică Legea lui Demeter și de ce?